

Topic C2: Elements, compounds and mixtures

C2.1 Purity and separating mixtures

Summary

In chemical terms elements and compounds are pure substances and mixtures are impure substances. Chemically pure substances can be identified using melting point. Many useful materials that we use today are mixtures. There are many methods of separating mixtures including filtration, crystallisation, distillation and chromatographic techniques.

Underlying knowledge and understanding

Learners should be familiar with the concept of pure substances. They should have met simple separation techniques of mixtures: filtration, evaporation and distillation. The identification of pure substances in terms of melting point, boiling point and chromatography will also have been met before.

Common misconceptions

Learners commonly misuse the word pure and confuse it with natural substance or a substance that has not been tampered with. They think that when a substance dissolves that the solution is pure and not a mixture.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
CM2.1i	arithmetic computation, ratio, percentage and multistep calculations permeates quantitative chemistry	M1a, M1c, M1d
CM2.1ii	provide answers to an appropriate number of significant figures	M2a
CM2.1iii	change the subject of a mathematical equation	M3b, M3c
CM2.1iv	arithmetic computation and ratio when determining empirical formulae, balancing equations	M3b, M3c

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
C2.1a	explain what is meant by the purity of a substance, distinguishing between the scientific and everyday use of the term 'pure'		WS1.4a	Purification of compounds. (PAG C4, PAG C7)
C2.1b	use melting point data to distinguish pure from impure substances	M1a, M1c, M1d, M2a		Measurement of melting point.
C2.1c	calculate relative formula masses of species separately and in a balanced chemical equation	M3b, M3c	WS1.3c, WS1.4c	
C2.1d	deduce the empirical formula of a compound from the relative numbers of atoms present or from a model or diagram and vice versa	M3b, M3c	WS1.1b, WS1.4a	
C2.1e	explain that many useful materials are formulations of mixtures	alloys		
C2.1f	describe, explain and exemplify the processes of filtration, crystallisation, simple distillation, and fractional distillation	knowledge of the techniques of filtration, crystallisation, simple distillation and fractional distillation	WS1.2b, WS1.2c, WS2a, WS2b	Separation of mixtures and purification of compounds. (PAG C4, PAG C7) Distillation of mixtures (PAG C4)
C2.1g	describe the techniques of paper and thin layer chromatography	using aqueous and non-aqueous solvents and locating agents	WS1.2b, WS1.2c, WS1.4a, WS2a, WS2b	Paper or thin layer chromatography. (PAG C3)
C2.1h	recall that chromatography involves a stationary and a mobile phase and that separation depends on the distribution between the phases	identification of the mobile and stationary phases	WS1.4a	

C2.1i	interpret chromatograms, including measuring R_f values	the recall and the use of the formula	M3b, M3c	WS1.3c, WS1.4a	
C2.1j	suggest suitable purification techniques given information about the substances involved				
C2.1k	suggest chromatographic methods for distinguishing pure from impure substances	paper, thin layer (TLC) and gas chromatography		WS1.4a	Using chromatography to identify mixtures of dyes in an unknown ink. (PAG C3)